

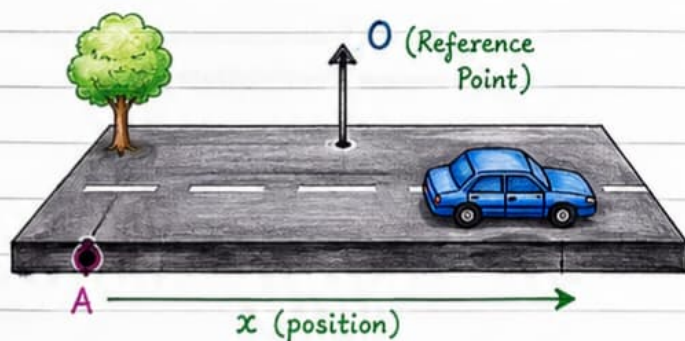
CHAPTER 2

MOTION IN A STRAIGHT LINE

“The branch of physics which deals with the motion of an object along a straight line is called Motion in a Straight Line.”

1. BASIC CONCEPTS

- Motion** : An object is said to be in motion if its position changes with time with respect to a reference point.
- Rest** : If there is no change in position with time, the object is at rest.
- Reference Point** : A point with respect to which the position of an object is measured.



2. SCALAR AND VECTOR QUANTITIES

SCALAR QUANTITY

- Magnitude only
- No direction
- Obeys ordinary rules of algebra

Examples : Distance, Speed, Time, Mass, Temperature.

VS

VECTOR QUANTITY

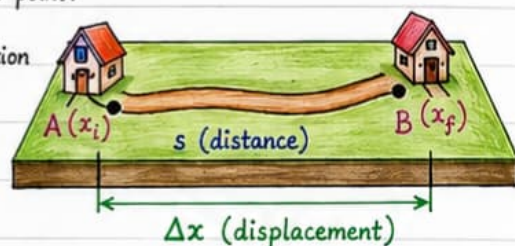
- Has both magnitude and direction
- Obeys vector laws of addition

Examples : Displacement, Velocity, Acceleration, Force.

3. POSITION, DISPLACEMENT AND DISTANCE

- Position (x)** : The location of an object with respect to a reference point.
- Displacement (Δx)** : The shortest distance between initial and final position

$$\Delta x = x_f - x_i$$
 [It is a vector]
- Distance (s)** : The actual path length travelled by the object
 [It is a scalar]



4. AVERAGE SPEED AND AVERAGE VELOCITY

Average Speed

$$= \frac{\text{Total Distance Travelled}}{\text{Total Time Taken}}$$

Average Velocity

$$= \frac{\text{Total Displacement}}{\text{Total Time Taken}}$$

Note : Average velocity can be zero even when average speed is not zero.



Key Idea : Displacement depends only on initial and final position, but distance depends on the path.



5. VELOCITY AND SPEED

- Speed** : Scalar quantity. It is the distance travelled per unit time.

$$\text{Speed (v)} = \frac{\text{Distance (s)}}{\text{Time (t)}}$$

SI unit : m s^{-1}

- Velocity ($\vec{v}$)** : Vector quantity. It is the displacement per unit time.

$$\text{Velocity ($\vec{v}$)} = \frac{\text{Displacement ($\Delta \vec{x}$)}}{\text{Time (t)}}$$

SI unit : m s^{-1}

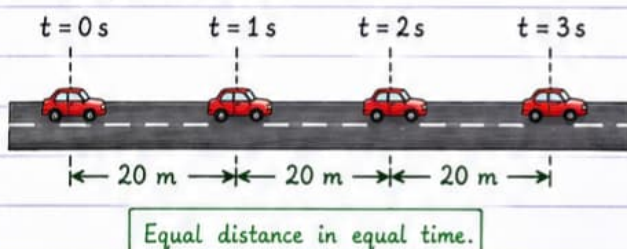
Remember!

- ★ Speed has only magnitude.
- ★ Velocity has magnitude and direction.
- ★ Average speed $\geq$ |Average velocity|

6. UNIFORM AND NON-UNIFORM MOTION

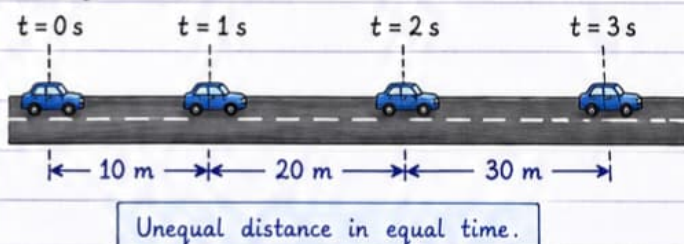
Uniform Motion

- An object covers equal distances in equal intervals of time along a straight line.
- Velocity (both magnitude and direction) remains constant.



Non-Uniform Motion

- An object covers unequal distances in equal intervals of time along a straight line.
- Velocity changes with time (change in magnitude or direction or both).



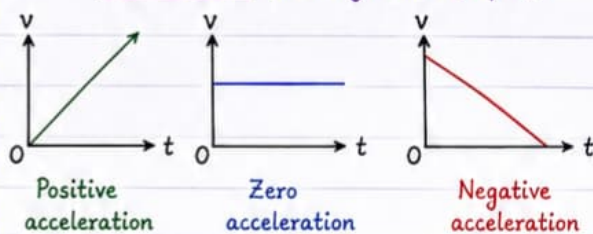
7. ACCELERATION

- Acceleration ($\vec{a}$)** : Rate of change of velocity of an object.

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i} \quad \text{SI unit : } \text{m s}^{-2}$$

- Positive acceleration** : Velocity increases with time.
- Negative acceleration (Retardation)** : Velocity decreases with time.
- Zero acceleration** : Velocity remains constant.

Visualization (Velocity-Time Graphs)



8. EQUATIONS OF MOTION (For Uniform Acceleration)

For motion in a straight line with constant acceleration (a), the following equations hold good :

- $v = u + at$ [First equation]
- $s = ut + \frac{1}{2} at^2$ [Second equation]
- $v^2 = u^2 + 2as$ [Third equation]

$\vec{u}$ = Initial velocity
 $\vec{v}$ = Final velocity
 $\vec{a}$ = Acceleration (constant)
 t = Time
 $\vec{s}$ = Displacement

Derivation (Idea) for Third Equation

$$\text{From } v = u + at \Rightarrow a = \frac{v-u}{t} \quad \dots (i)$$

$$\text{Put this in } s = ut + \frac{1}{2} at^2$$

$$s = ut + \frac{1}{2} \left(\frac{v-u}{t} \right) t^2$$

$$s = ut + \frac{1}{2} (v-u) t \quad \dots (ii)$$

$$\text{Again from (i), } t = \frac{v-u}{a}$$

$$\text{Put in (ii) and simplify } \Rightarrow v^2 = u^2 + 2as$$

Note :

These equations are valid only for constant acceleration.

Key Point : In straight line motion, everything depends on how position or velocity changes with time.

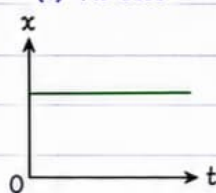
9. GRAPHICAL REPRESENTATION OF MOTION

9.1 Position-Time Graph (x-t Graph)

- It shows how position of an object changes with time.
- The slope of x-t graph gives velocity.

$$\text{Slope} = \frac{\Delta x}{\Delta t} = \text{velocity}$$

(i) At rest



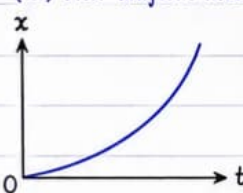
$$\text{Slope} = 0 \\ v = 0$$

(ii) Uniform motion



$$\text{Slope} = \text{constant} \\ v = \text{constant}$$

(iii) Non-uniform motion



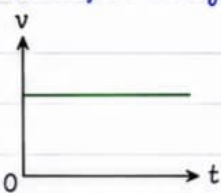
$$\text{Slope} = \text{changing} \\ v = \text{changing}$$

9.2 Velocity-Time Graph (v-t Graph)

- It shows how velocity of an object changes with time.
- The slope of v-t graph gives acceleration.

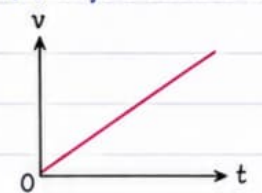
$$\text{Slope} = \frac{\Delta v}{\Delta t} = \text{acceleration}$$

(i) Uniform velocity



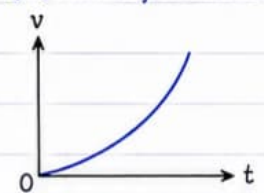
$$\text{Slope} = 0 \\ a = 0$$

(ii) Uniform acceleration



$$\text{Slope} = \text{constant} \\ a = \text{constant}$$

(iii) Non-uniform acceleration

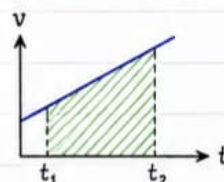


$$\text{Slope} = \text{changing} \\ a = \text{changing}$$

Key Idea :

Area under v-t graph between two instants gives displacement.

Displacement (s) = Area under v-t graph
= Area of shaded region

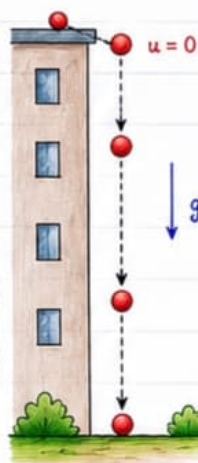


10. SPECIAL CASES OF UNIFORMLY ACCELERATED MOTION

10.1 Free Fall

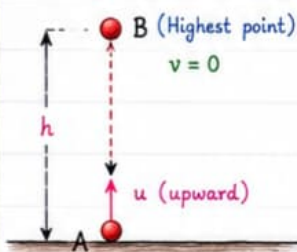
- When an object is dropped from rest and allowed to fall freely under gravity.
- Acceleration due to gravity = $g \approx 9.8 \text{ ms}^{-2}$ (downward)

$u = 0$, $a = g$, directed vertically downward.
Equations of motion can be used by putting $a = g$.



10.2 Projection of a Body Vertically Upwards

- A body thrown vertically upward has initial velocity u .
- Acceleration is g downward ($-g$).
- At the highest point, final velocity $v = 0$.



At highest point

- Velocity = 0
- Acceleration = g (downward)

During upward motion : $a = -g$
During downward motion : $a = +g$

11. RELATION BETWEEN DIFFERENT QUANTITIES

- If acceleration is zero ($a = 0$) :
 $v = u$ (velocity is constant)
- If velocity is zero ($v = 0$) :
Object momentarily comes to rest.
- If displacement is zero ($s = 0$) :
Object returns to the initial position.
- If distance $\neq$ displacement :
Path is curved or object changes direction.

Important Point

All equations of motion are valid only for motion in a straight line with constant acceleration.

Remember : Sign of acceleration depends on the chosen positive direction.

DIMENSIONS (For NEET)

- $[x] = L$
- $[v] = L T^{-1}$
- $[a] = L T^{-2}$
- $[s] = L$
- $[t] = T$

NEET Important

Conceptual understanding of graphs and equations is essential for solving problems quickly in NEET.

Summary : In straight line motion, position, velocity and acceleration are related through equations of motion and graphs.

12. NUMERICALS (BASED ON NCERT EXAMPLES)

Example 1 A train travels a distance of 360 km in 6 hours. Find its average speed.

Solution :

Distance (s) = 360 km

Time (t) = 6 h

$$\text{Average speed} = \frac{s}{t} = \frac{360}{6} = 60 \text{ km h}^{-1}$$



Example 3 A body starts from rest and moves with uniform acceleration 3 m s^{-2} for 5 s.

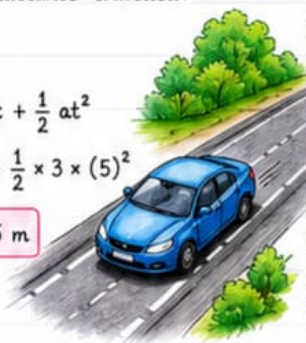
Find (i) final velocity (ii) distance travelled.

Solution :

$$\begin{aligned} \text{(i) } v &= u + at \\ &= 0 + 3 \times 5 \end{aligned}$$

$$v = 15 \text{ m s}^{-1}$$

$$\begin{aligned} \text{(ii) } s &= ut + \frac{1}{2}at^2 \\ &= 0 + \frac{1}{2} \times 3 \times (5)^2 \\ &= 37.5 \text{ m} \end{aligned}$$



Example 2 A car accelerates uniformly from 0 to 72 km h^{-1} in 10 s. Find acceleration.

Solution :

$$u = 0$$

$$v = 72 \text{ km h}^{-1} = 72 \times \frac{5}{18} = 20 \text{ m s}^{-1}$$

$$t = 10 \text{ s}$$

$$\begin{aligned} a &= \frac{v - u}{t} \\ &= \frac{20 - 0}{10} = 2 \text{ m s}^{-2} \end{aligned}$$

Example 4 A ball is thrown vertically upward with speed 20 m s^{-1} . Find (i) time to reach the highest point (ii) maximum height reached.

Solution :

(i) At highest point, $v = 0$

$$v = u + at \Rightarrow 0 = 20 + at$$

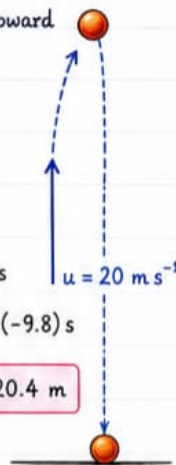
$$a = -g = -9.8 \text{ m s}^{-2}$$

$$t = \frac{20}{9.8} = 2.04 \text{ s}$$

(ii) $v^2 = u^2 + 2as$

$$0 = (20)^2 + 2(-9.8)s$$

$$s = \frac{400}{19.6} = 20.4 \text{ m}$$



13. IMPORTANT FORMULAS AT A GLANCE

Quantity	Formula	Remarks
• Average Speed	$\bar{v} = \frac{\text{Total distance}}{\text{Total time}}$	Scalar quantity
• Average Velocity	$\bar{v} = \frac{\text{Total displacement}}{\text{Total time}}$	Vector quantity
• Velocity	$v = u + at$	First equation of motion
• Displacement	$s = ut + \frac{1}{2}at^2$	Second equation of motion
• Relation	$v^2 = u^2 + 2as$	Third equation of motion
• Acceleration	$a = \frac{v - u}{t}$	Rate of change of velocity
• Speed	$v = \frac{s}{t}$	Scalar speed

14. DIMENSIONAL ANALYSIS

- Displacement (s) $\rightarrow [L]$
- Velocity (v) $\rightarrow [LT^{-1}]$
- Acceleration (a) $\rightarrow [LT^{-2}]$
- Time (t) $\rightarrow [T]$

All equations of motion are dimensionally correct.



NEET KNOW

In NEET problems, questions often involve change in velocity, time taken, and distance covered using equations of motion and graphs.



15. SUMMARY

- Motion in a straight line is described using position, displacement and distance.
- Speed is scalar, velocity is vector.
- For uniform motion, velocity is constant; for non-uniform motion, it changes.
- Acceleration is the rate of change of velocity.
- Equations of motion (for constant acceleration) relate u , v , a , t and s .
- Graphs (x - t , v - t) help in understanding motion visually.
- Sign convention is important: choose positive direction and follow consistently.
- These concepts are the foundation for further topics in physics.

16. SI UNITS

- Length $\rightarrow$ metre (m)
- Time $\rightarrow$ second (s)
- Speed $\rightarrow \text{m s}^{-1}$
- Velocity $\rightarrow \text{m s}^{-1}$
- Acceleration $\rightarrow \text{m s}^{-2}$



Key Takeaway : Understanding straight line motion deeply helps you solve many real-world and exam-level problems with confidence.